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INSCONE: Unknown-Aware Detection of LLM-Generated Text via Informed Wild Data

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Abstract

AI-generated text is increasingly difficult to distinguish from human writing, creating risks in academic integrity, medical misinformation, and social media disinformation. While recent work has reframed MGT detection as an out-of-distribution problem (Zeng et al., 2025) with strong in-distribution and zero-shot results, generalization to LLMs unseen during training remains underexplored. We address this with two contributions. First, we propose **INSCONE** (**In**formed **SCONE**), which adapts the SCONE wild-data framework (Bai et al., 2025) to the text domain by exploiting curated wild data with known mixing proportions to stabilize the energy geometry around seen and unseen LLM families, achieving strong zero-shot generalization on the RAID benchmark (Dugan et al., 2024). Second, we extend the RAID benchmark with regenerations from contemporary frontier models, providing a broader empirical picture of detector behavior under realistic deployment conditions. Code and data are available at `github.com/markstanl/INSCONE` and `huggingface.co/datasets/markstanl/RAID-Plus`.

Keywords: machine-generated text detection, out-of-distribution detection, open-set recognition, wild data, energy-based models

1. Introduction

The core challenge for any practical MGT detector is *unknown-awareness*: reliably identifying AI-generated text from model families never seen during training. Pu et al. (2022) quantify this directly — no single supervised detector generalizes across LLM families, and GPT and LLaMA families are largely isolated from each other in detector space. This ceiling is fundamental, not incidental: as LLMs proliferate, retraining a detector for each new model family is not a viable deployment strategy.

Zeng et al. (2025) reframe detection as open-set recognition, treating LLM-generated text as in-distribution and human writing as semantic OOD. Under this framing, unseen LLMs are covariate OOD: same semantic class as ID machine text, but shifted distribution. An energy-based detector draws a boundary around known AI-generated text and flags anything outside as human or unknown. This formulation generalizes better than supervised classifiers because the decision surface is anchored to the ID region rather than learned boundaries between human and machine text.

The natural extension is wild data: unlabeled mixtures of covariate and semantic OOD that improve generalization to unseen LLMs, following Bai et al. (2025). We identify a fundamental mismatch, however. SCONE's uniform wild push is benign in vision because covariate shift is pixel-level and Lipschitz drag from ID samples dominates. In text, covariate shift means an entirely different model family — embedding distance from ID is large and heterogeneous — so the push term overwhelms the drag and incorrectly drives unseen LLM embeddings toward the human text region.

Our key insight is that when the wild batch is explicitly curated with known mixing proportions $(\pi_{\mathrm{id}}, \pi_c, \pi_s)$, the quantile boundary between covariate and semantic OOD is exact and requires no estimation. We call this **INSCONE** (**In**formed **SCONE**) and implement it as a proximal/distal energy split with a dead-zone buffer. INSCONE improves zero-shot generalization without architectural changes, and without retraining for each new model family. We further release **RAID+**, an extended evaluation set regenerating RAID prompts with frontier models not present in the original benchmark.

Our contributions are: **(1)** INSCONE, an informed wild-data loss that resolves the covariate/semantic conflation of SCONE for text; and **(2)** RAID+, an extended evaluation set regenerating RAID prompts with contemporary frontier models unavailable at the time of the original benchmark's publication.

2. Related Work

MGT detection. Zero-shot methods (Mitchell et al., 2023; Hans et al., 2024) and supervised classifiers (Hu et al., 2023; Verma et al., 2024) both assume a fixed model landscape. Pu et al. (2022) train ELECTRA-large classifiers on 13 LLMs and find that detectors trained on medium-sized models generalize within a family but not across families. Wu et al. (2024) identify cross-model generalization as the central open problem.

OOD detection for MGT. Zeng et al. (2025) reformulate MGT detection as OSR, combining DeepSVDD, HRN, and energy-based scoring with contrastive pretraining on SimCSE-RoBERTa. The key insight is that LLM-generated text occupies a tighter distributional region than human text, making LLM text the natural in-distribution class. Our work builds directly on their energy-based variant.

Wild data and open-set generalization. Bai et al. (2025) propose SCONE for vision OSR, using unlabeled wild data with an energy margin loss. The theoretical justification relies on a Lipschitz drag argument (Proposition 3.1 of Bai et al. 2025) that assumes small covariate shifts, an assumption violated in the text domain. du Plessis et al. (2015) establish that known class priors enable strictly better label assignment to unlabeled data in PU learning, which motivates our informed quantile split.

3. Methods

3.1. Baseline: Energy-Based MGT Detector

Following Zeng et al. (2025), we use a frozen SimCSE-RoBERTa encoder (Gao et al., 2022) (`princeton-nlp/unsup-simcse-roberta-base`) with a 3-layer MLP classification head over K LLM family classes. The free energy score $E(x) = -\log \sum_i \exp(f_i(x))$ serves as the anomaly score: low energy indicates a confident ID (machine) prediction; high energy flags human or unseen LLM text.

The combined training loss is:

$$\mathcal{L} = \alpha_{\text{con}}\mathcal{L}_{\text{con}} + \alpha_{\text{cls}}\mathcal{L}_{\text{cls}} + \alpha_{\text{eng}}\mathcal{L}_{\text{eng}} \tag{1}$$

where $\mathcal{L}_{\text{con}}$ is a SimCLR-style binary contrastive loss grouping machine vs. human text, $\mathcal{L}_{\text{cls}}$ is cross-entropy over known LLM families on machine text only, and

$$\mathcal{L}_{\text{eng}} = \mathbb{E}\big[\text{ReLU}(E - m_{\text{in}})^2 \cdot \mathbf{1}_{y=1} + \text{ReLU}(m_{\text{out}} - E)^2 \cdot \mathbf{1}_{y=0}\big] \tag{2}$$

shapes the energy surface with margins $m_{\text{in}} = -7.0$ (ID anchor) and $m_{\text{out}} = -2.0$ (OOD anchor). At inference, a threshold τ at the 95th percentile of ID validation energy calibrates the FPR95 operating point.

3.2. INSCONE: Informed Wild-Data Loss

Problem setup. To simulate wild data without requiring internet-scraped unlabeled corpora, we use held-out LLMs. Our data is split into three categories: ID models used during standard training, OOD-covariate models used with the wild loss, and unseen LLMs held out entirely for zero-shot testing. We choose these models to balance a temporal scale while maintaining at least one model per family for energy classifier coverage. Critically, because the wild batch is explicitly curated, the mixing proportions $(\pi_{\text{id}}, \pi_c, \pi_s)$ are known exactly, and the quantile boundary between covariate and semantic OOD requires no estimation.

Why standard SCONE is insufficient for text. The original SCONE wild loss $\text{ReLU}(-E_{\text{wild}})^2$ pushes all wild samples toward high energy. In vision OSR, this is harmless: covariate shift is pixel-level, and the Lipschitz drag from ID samples pulls nearby covariate samples back down (Bai et al., 2025). In text, covariate shift means a different model family entirely. The embedding distance from ID is large and heterogeneous, so the push term dominates and covariate LLMs are incorrectly driven toward the human text region. Empirically, the FPR95 of wild data degrades from $0.1942 \to 0.3848$ under standard SCONE relative to the data-matched baseline.

Informed quantile split. Given known mixing proportions, we define proximal and distal thresholds:

$$\tau_{\text{prox}} = \text{Quantile}(E_{\text{wild}},\ 1 - \pi_s - \delta/2) \tag{3}$$
$$\tau_{\text{dist}} = \text{Quantile}(E_{\text{wild}},\ 1 - \pi_s + \delta/2) \tag{4}$$

where δ is a buffer parameter that creates a dead zone around the composition boundary. The INSCONE loss is:

$$\begin{aligned}\mathcal{L}_{\text{INSCONE}} = {} & \text{ReLU}(E_{\text{wild}}[E \leq \tau_{\text{prox}}] - m_{\text{in}})^2 \\ & + \text{ReLU}(-E_{\text{wild}}[E \geq \tau_{\text{dist}}])^2\end{aligned} \tag{5}$$

The proximal term pulls estimated covariate OOD toward the ID energy region; the distal term pushes estimated semantic OOD toward high energy. Samples in $(\tau_{\text{prox}}, \tau_{\text{dist}})$ receive no gradient, preventing spurious updates from compositionally ambiguous samples. At

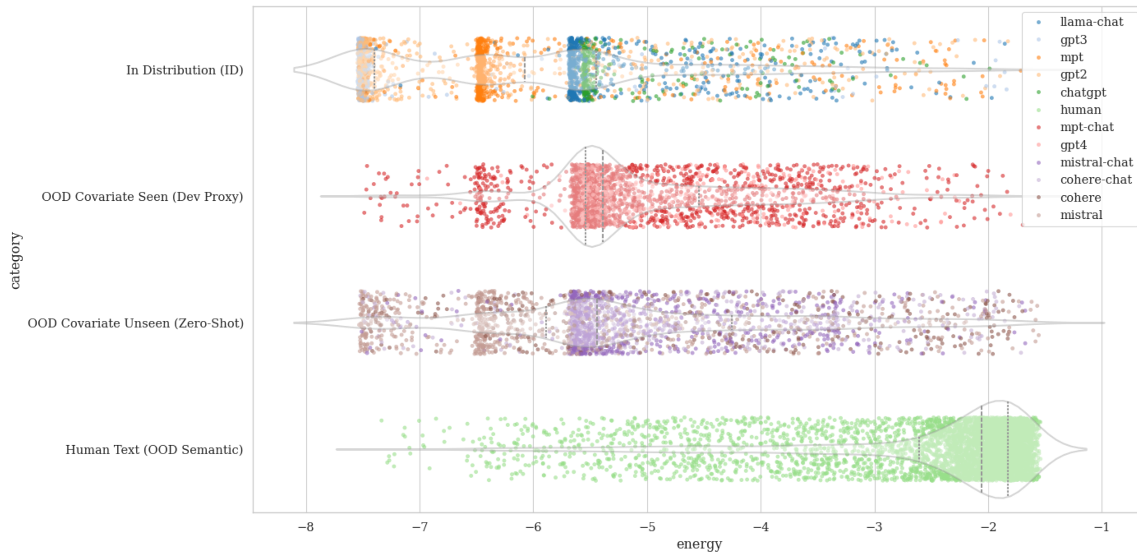

Figure 1: Energy distributions by split under the baseline trained on only ID and human data. ID and covariate LLMs occupy similar low-energy regions; human text concentrates near the OOD margin. Zero-shot unseen models track the covariate seen distribution, supporting the temporal generalization hypothesis.

$\delta = 0$, this recovers a clean π_s-split. Because this relies on the wild energy distribution being ordered as described, we apply a warmup of several epochs before activating the wild loss. The total loss adds $\lambda_{\text{INSCONE}}\mathcal{L}_{\text{INSCONE}}$ to Equation (1).

This approach relies on the assumption that unseen LLM data occupies lower energy than human text under the trained baseline. Figure 1 confirms this empirically: zero-shot LLMs track the energy distribution of covariate seen models rather than human text. Figure 2 corroborates this at the wild batch level: the quantile split cleanly partitions by class, with $\sim$99.2% of proximal samples being ID or covariate OOD and $\sim$99.6% of distal samples being human text.

Finally, while our formulation simulates wild data via held-out LLMs, the approach extends naturally to truly unlabeled corpora: a reasonable estimate of $(\pi_{\text{id}}, \pi_c, \pi_s)$ combined with the buffer δ is sufficient to apply the proximal/distal split, making INSCONE applicable beyond the controlled experimental setting.

4. Experimental Setup

Dataset. We use the RAID dataset (Dugan et al., 2024) for all experiments. Our **scone-temporal split** partitions models by approximate release date: ID = {GPT-2, GPT-3, MPT, ChatGPT, LLaMA-Chat}; covariate OOD = {GPT-4, MPT-Chat}; zero-shot test = {Mistral, Mistral-Chat, Cohere, Cohere-Chat}. Human text is semantic OOD throughout. Due to compute constraints, all experiments are conducted on a held-out subset of RAID

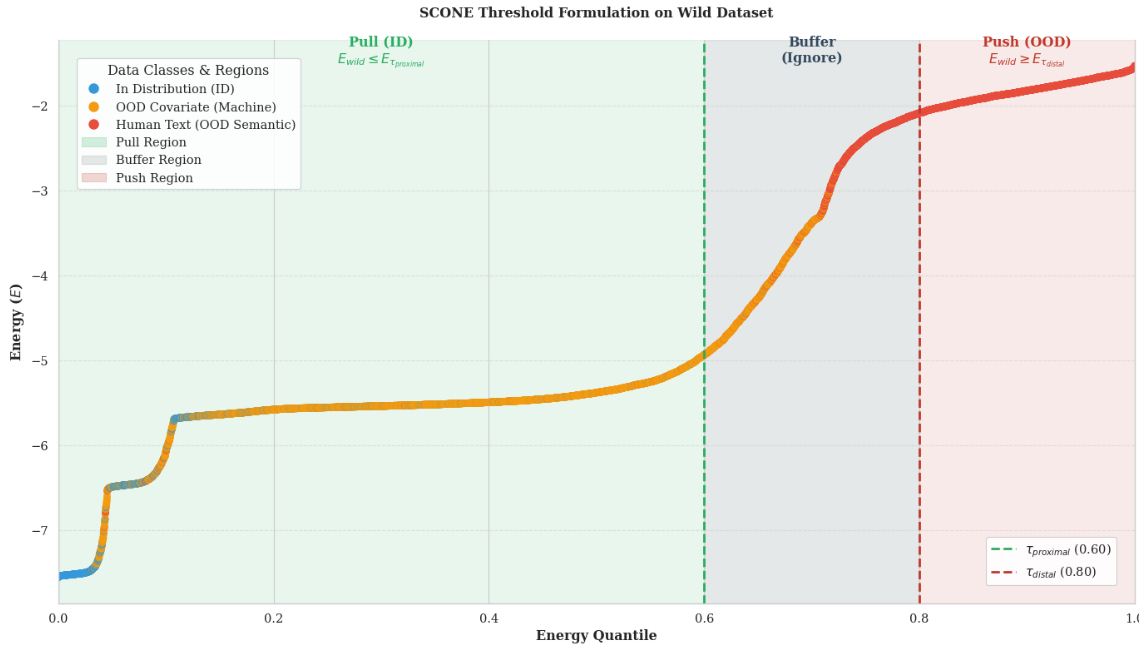

Figure 2: Empirical wild batch energy quantile curve with wild ratios $(0.1, 0.6, 0.3)$ and buffer δ=0.1. The proximal region (green) anchors covariate machine samples at low energy; the distal region (red) anchors human text at high energy; the buffer (gray) receives no gradient. ∼99.2% of proximal samples are ID or covariate OOD, and ∼99.6% of distal samples are human text.

rather than the full benchmark. We additionally construct **RAID+** by regenerating RAID prompts with frontier models not available at HTAO publication: Gemini-3.1-Pro, DeepSeek-V3, Gemma-3-27B, and LLaMA-3.3-70B ($\approx$ 2,000 samples per model, evaluation only).

Compared methods. All variants share the same encoder: SimCSE-RoBERTa-base with intermediate-layer feature fusion (layers 6 and 12), and a 3-layer MLP classification head ($768 \to 192 \to 48 \to$ K).

- **Baseline**: no wild data, λ_{INSCONE}=0.
- **Baseline (fair)**: covariate seen models (GPT-4, MPT-Chat) added to labeled ID training, controlling for data access.
- **SCONE**: identical setup to INSCONE but using the original uniform wild push loss (Bai et al., 2025).
- **INSCONE (ours)**: λ_{INSCONE}=0.005, δ=0.1, $\pi = (0.1, 0.6, 0.3)$.

Training. Training uses AdamW with a cosine learning rate schedule, `bf16-mixed` precision, and a 500-step linear warmup. The SCONE wild loss is delayed by 4 epochs to allow

Table 1: Main results on RAID temporal split. INSCONE achieves the best zero-shot FPR95.

Method	Overall AUROC ↑	Wild FPR95 ↓	Zero-Shot FPR95 ↓
Baseline	0.9577	0.1646	0.2862
Baseline (fair)	0.9571	0.1552	0.3166
SCONE	0.9164	0.3848	0.4270
INSCONE (ours)	0.9568	0.1764	**0.2252**

the energy surface to stabilize before wild gradients activate. Key hyperparameters are m_{in}=-7.0, m_{out}=-2.0, and α_{eng}=0.001. Checkpoints are selected by the mean of id and wild AUROC.

Metrics. AUROC and FPR95 (false positive rate at 95% TPR) on the wild and zero shot evaluation pillars.

5. Results

Table 1 reports results on the full-scale RAID scone-temporal split. INSCONE achieves the best zero-shot FPR95 (0.2252), a 6.1-point improvement over the naive baseline and a 20.3-point improvement over standard SCONE. This comes at a modest cost: wild FPR95 (0.1764) trails the fair baseline (0.1552) by 2.1 points, and overall AUROC (0.9568) is within 0.001 of both baselines. We interpret this as the expected tradeoff of the proximal pull, redirecting some covariate pressure toward zero-shot generalization necessarily relaxes the fit on seen covariate models.

Two findings are worth highlighting. First, the fair baseline does not improve zero-shot FPR95 and slightly degrades it (0.3166 vs. 0.2862). This is consistent with Pu et al. (2022): adding more labeled data causes the classifier to memorize seen families rather than generalize. INSCONE avoids this by exposing covariate models only through the wild loss, without assigning a family label. Second, standard SCONE is strictly worse than the naive baseline on all three metrics, confirming that the uniform push formulation is not appropriate for the text domain.

Ablation. Table 2 reports a sweep over wild ratios and buffer size (3 seeds), used to select the best configuration. The best zero-shot FPR95 (0.2991) is achieved at $\pi = (0.1, 0.6, 0.3)$, $\delta = 0.1$. At $\delta = 0.0$, boundary samples receive gradient and introduce noise. At $\delta = 0.2$, the dead zone discards too many informative samples. Removing the semantic OOD component ($\pi = (0.0, 1.0, 0.0)$) is the worst configuration, confirming the distal push on human text is a necessary component. Despite high run variance across seeds, all INSCONE configurations improve zero-shot FPR95 over baseline, suggesting the gain is robust to hyperparameter choice.

Table 2: Hyperparameter ablation (3 seeds). δ=0.1 with $\pi = (0.1, 0.6, 0.3)$ is the best configuration. Removing semantic OOD (π_s=0) collapses performance.

Ratios (π)	δ	**AUROC** ↑	**Wild** ↓	**ZS** ↓
Baseline (no wild)		0.9175	0.6094	0.5860
Baseline (fair)		0.9224	0.1942	0.4228
$(0.1, 0.6, 0.3)$	0.0	0.9345	0.2265	0.3166
$(0.1, 0.6, 0.3)$	0.1	**0.9400**	**0.1827**	**0.2991**
$(0.1, 0.6, 0.3)$	0.2	0.9406	0.2506	0.3802
$(0.0, 0.7, 0.3)$	0.0	0.9316	0.2440	0.3188
$(0.0, 1.0, 0.0)$	0.0	0.9116	0.3337	0.4142

6. Conclusion

We introduced INSCONE, an informed wild-data loss that resolves a fundamental mismatch between SCONE's vision-domain assumptions and the structure of LLM embedding space. By exploiting known wild batch composition, INSCONE applies targeted proximal pulls and distal pushes rather than uniform energy pushing, achieving the best zero-shot FPR95 (0.2252) across all compared methods while maintaining strong overall AUROC. The fair baseline comparison confirms that the wild-loss formulation — not additional data access — is the source of INSCONE's gains. We additionally released RAID+, an extended evaluation set covering frontier models not present in the original RAID benchmark. Future directions include adversarial evaluation of INSCONE and evaluation on RAID+ to characterize detector behavior against contemporary frontier models.

Limitations

INSCONE assumes that wild batch composition $(\pi_{\text{id}}, \pi_c, \pi_s)$ is known or approximately known at training time, which requires deliberate data curation or reliable estimation of wild data composition. In truly open-world settings where the fraction of human versus machine text is unknown, the proximal/distal split degrades.

Due to compute constraints, all experiments are conducted on a 20k subset of RAID rather than the full benchmark, and evaluation is limited to a single dataset. Whether the gains generalize to other MGT benchmarks (M4, TuringBench) or to non-English text remains untested. The encoder is also fixed at SimCSE-RoBERTa-base; stronger backbones with larger context windows or explicit family-level pretraining may shift the relative performance of the compared methods.

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Table 3: Training hyperparameters for the full-scale INSCONE run.

Hyperparameter	Value
Encoder	princeton-nlp/unsup-simcse-roberta-base
Head dims	768 $\rightarrow$ 192 $\rightarrow$ 48 $\rightarrow$ K
m_{in}	-7.0
m_{out}	-2.0
α_{con}	1.0
α_{cls}	1.0
α_{eng}	0.001
λ_{INSCONE}	0.005
Buffer δ	0.1
Wild ratios π	$(0.1, 0.6, 0.3)$
SCONE warmup	4 epochs
LR schedule	cosine, warmup 500 steps
Optimizer	AdamW
Precision	bf16-mixed

Appendix A. Hyperparameter Details

Table 3 lists all hyperparameters used for the full-scale INSCONE run reported in Table 1.

Appendix B. RAID Temporal Split Details

Table 4 details the model assignments for each partition of the scone-temporal split used in all experiments.

Table 4: RAID scone-temporal split. All experiments use 10,000 labeled training samples and 10,000 wild samples, besides the fair baseline setup, which uses 20,000 labeled training samples and 0 wild samples.

Split	Models
ID (train)	GPT-2, GPT-3, MPT, ChatGPT, LLaMA-Chat
Covariate OOD	GPT-4, MPT-Chat
Zero-shot (test)	Mistral, Mistral-Chat, Cohere, Cohere-Chat
Semantic OOD	Human text (all splits)

Appendix C. Per-Seed Ablation Results

Table 5 provides the full per-seed breakdown of the ablation summarized in Table 2, included to document run variance across configurations. High variance on many seeds.

Table 5: Per-seed ablation results. ZS = zero-shot FPR95, Wild = wild_memorization FPR95, AUC = AUROC.

		Seed 42			Seed 43			Seed 44		
Ratios (π)	δ	**ZS**	**Wild**	**AUC**	**ZS**	**Wild**	**AUC**	**ZS**	**Wild**	**AUC**
Baseline (no wild)		0.416	0.698	0.916	0.798	0.874	0.903	0.544	0.256	0.934
Baseline (fair)		0.577	0.205	0.923	0.349	0.182	0.927	0.343	0.195	0.918
$(0.1, 0.6, 0.3)$	0.0	0.275	0.166	0.951	0.346	0.330	0.944	0.328	0.183	0.909
$(0.1, 0.6, 0.3)$	0.1	0.243	0.139	0.944	0.340	0.233	0.943	0.314	0.176	0.934
$(0.1, 0.6, 0.3)$	0.2	0.257	0.189	0.939	0.277	0.174	0.951	0.606	0.389	0.932
$(0.0, 0.7, 0.3)$	0.0	0.243	0.190	0.944	0.386	0.359	0.921	0.327	0.183	0.930
$(0.0, 1.0, 0.0)$	0.0	0.441	0.480	0.922	0.461	0.321	0.896	0.341	0.200	0.917